Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A triangular lamina of area A and height h is immersed in a liquid of density ρ in a vertical plane with its base on the surface of the liquid. The thrust on the lamina is

(A) $\frac{1}{2}A\rho gh$

(B) $\frac{1}{3}A\rho gh$

(C) $\frac{1}{6}A\rho gh$

(D) $\frac{2}{3}A\rho gh$

Q.2

The velocity of efflux when 75% of the liquid has drained out is

(A) $\sqrt{\frac{gh}{2}}$

(B) $\sqrt{gh}$

(C) $\sqrt{\frac{3gh}{4}}$

(D) $\sqrt{\frac{3gh}{2}}$

Q.3

A block of wood floats in a liquid with four-fifths of its volume submerged. If the relative density of wood is 0.8, what is the the density of the liquid in units of kgm^{-3}?

(A) 750

(B) 1000

(C) 1250

(D) 1500

Q.4

The cylindrical tube of a spray pump has a radius R, one end of which has n fine holes, each of radius r. If the speed of flow of the liquid in the tube is V, the speed of ejection of the liquid through the holes is

(A) $\frac{V}{n}\left(\frac{R}{r}\right)^{1/2}$

(B) $\frac{V}{n}\left(\frac{R}{r}\right)$

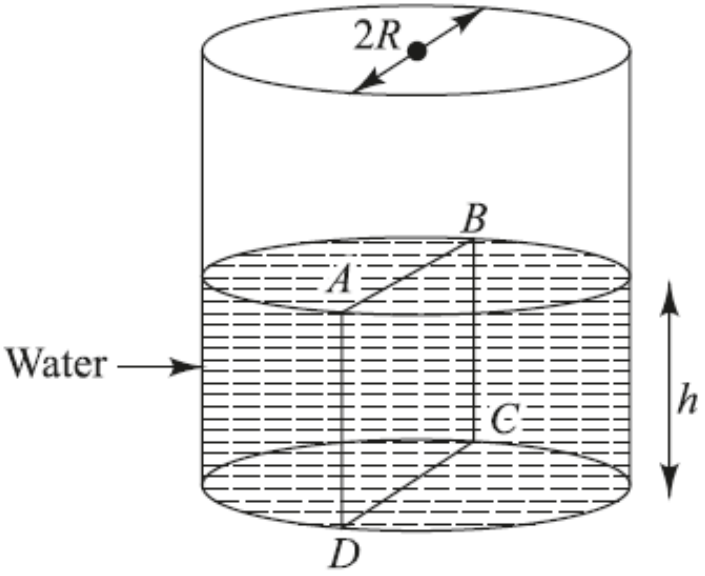
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(C) $\frac{V}{n}\left(\frac{R}{r}\right)^{3/2}$

(D) $\frac{V}{n}\left(\frac{R}{r}\right)^{2}$

Q.5

Water is filled up to a height h in a beaker of radius R as shown in the figure. The density of water is p, the surface tension of water is T and the atmospheric pressure is P_0. Consider a vertical section $ABCD$ of the water column through a diameter of the beaker. The force on water on one side of this section by water on the other side of this section has magnitude



(A) $\left|2P_0Rh+\pi R^2\rho gh-2RT\right|$

(B) $\left|2P_0Rh+R\rho gh^2-2RT\right|$

(C) $\left|P_0\pi R^2+R\rho gh^2-2RT\right|$

(D) $\left|P_0\pi R^2+R\rho gh^2+2RT\right|$

Q.6

A liquid flows through a pipe of varying diameter. The velocity of the liquid is $2\ \text{ms}^{-1}$ at a point where the diameter is $6\ \text{cm}$. The velocity of the liquid at a point where the diameter is $3\ \text{cm}$ will be

(A) $1\ \text{ms}^{-1}$

(B) $4\ \text{ms}^{-1}$

(C) $8\ \text{ms}^{-1}$

(D) $16\ \text{ms}^{-1}$

Q.7

A closed compartment containing gas is moving with some acceleration in horizontal direction. Neglect the effect of gravity. Then the pressure in the compartment is

(A) the same everywhere

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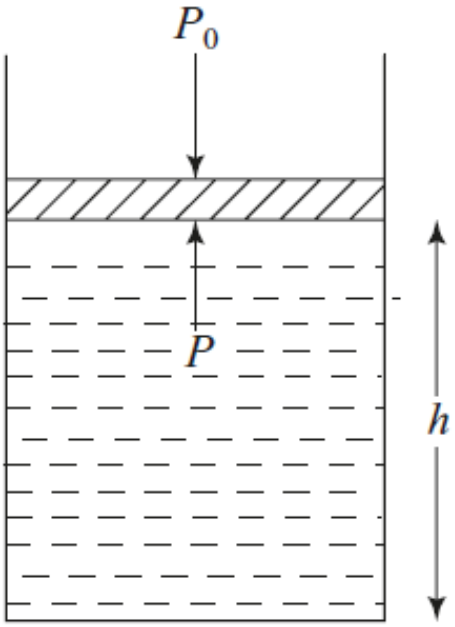
(B) lower in the front side

(C) lower in the rear side

(D) lower in the upper side

Q.8

A cylindrical vessel is filled to a height h by a liquid of density ρ. A tight-fitting frictionless piston of mass m is placed on the liquid as shown in Fig. 7.4. If P_0 is the atmospheric pressure and a the crosssectional area of the piston, the pressure P of the liquid just below the piston is



(A) P_0

(B) $P_0 - \frac{mg}{a}$

(C) $P_0 + \frac{mg}{a}$

(D) $P_0 - h\rho g + \frac{mg}{a}$

Q.9

A water barrel having water upto a depth d is placed on a table of height h. A small hole is made on the wall of the barrel at its bottom. If the stream of water coming out of the hole falls on the ground at a horizontal distance ' R ' from the barrel, then the value of d is

(A) $\frac{4h^2}{R}$

(B) $\frac{2h^2}{R}$

(C) $\frac{R^2}{4h}$

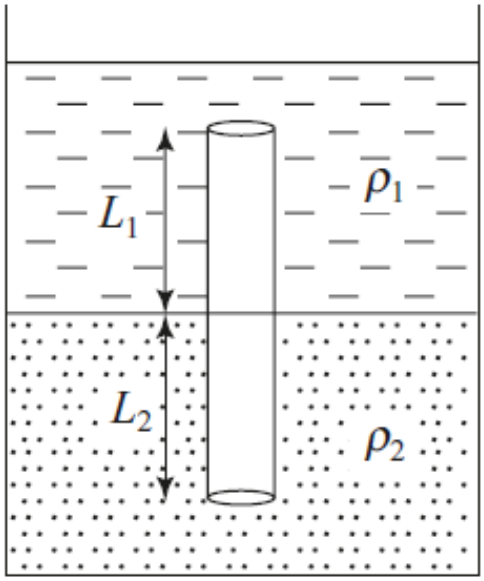
(D) $\frac{R^2}{2h}$

Q.10

A uniform cylinder of density σ and length L floats vertically completely immersed in two liquids with length

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L_1 immersed in a liquid of density ρ_1 and the remaining length L_2 immersed in the other liquid of density ρ_2 as shown in Fig. 7.12. The ratio $\frac{L_1}{L_2}$ will be $(\rho_1 > \rho_2)$ equal to



(A) $\frac{\sigma-\rho_2}{\sigma-\rho_1}$

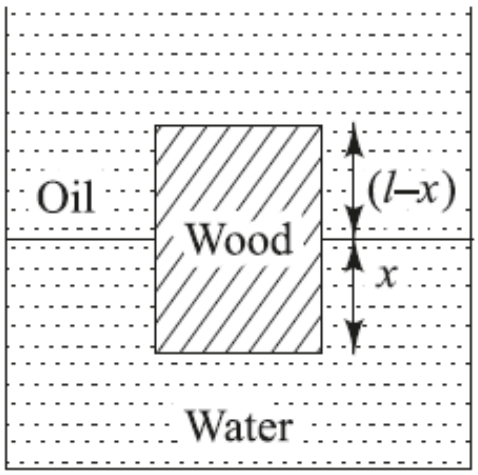
(B) $\frac{\sigma-\rho_1}{\rho_2-\sigma}$

(C) $\frac{\sigma-\rho_2}{\rho_1-\sigma}$

(D) $\frac{\sigma}{\rho_1-\rho_2}$

Q.11

A cubical block of wood of side l floats at the interface between oil of relative density n and water with its lower face $x = l/4$ below the interfaces as shown in Fig. The relative density of wood is



(A) $\frac{1}{2}(n+1)$

(B) $\frac{1}{3}(2n+1)$

(C) $\frac{1}{4}(3n+1)$

(D) $(1-n)$

Q.12

The excess pressure across a soap bubble of radius r is $p = 4\sigma/r$, where σ is the surface tension of soap

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solution. What is the excess pressure across an air bubble of the same radius r formed inside a container of soap solution?

(A) $\frac{\sigma}{r}$

(B) $\frac{2\sigma}{r}$

(C) $\frac{4\sigma}{r}$

(D) $\frac{2\sqrt{2}\sigma}{r}$

Q.13

The work done to break up a drop of a liquid of radius R and surface tension σ into eight drops, all of the same size, is

(A) $4\pi\sigma R^2$

(B) $2\pi\sigma R^2$

(C) $\frac{1}{2}\pi\sigma R^2$

(D) $\frac{1}{4}\pi\sigma R^2$

Q.14

A small metal sphere of radius r and density ρ falls from rest in a viscous liquid of density σ and coefficient of viscosity η. Due to friction heat is produced. The rate of production of heat when the sphere has acquired the terminal velocity is proportional to

(A) r^2

(B) r^3

(C) r^4

(D) r^5

Q.15

A small steel ball of radius r experiences a viscous force F when it is falling in a jar of glycerine with terminal velocity v. The viscons force experienced by a steel ball of radius $r/2$ falling in glycerine with terminal velocity $v/2$ is

(A) F

(B) $\frac{F}{2}$

(C) $\frac{F}{4}$

(D) $\frac{F}{8}$

Q.16

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A capillary tube of radius r is immersed in water and water rises in it to a height h. The mass of water in the capillary tube is 5 g. Another capillary tube of radius $2r$ is immersed in water. The mass of water that will rise in this tube is

(A) 2.5 g

(B) 5.0 g

(C) 10 g

(D) 20 g

Q.17

A cylindrical vessel of crosssectional area A is filled with a liquid to a height h and placed on a horizontal surface. The vessel has a small hole of cross-sectional area a on its side near the bottom. The hole is unplugged. If $a << A$, the minimum coefficient of friction between the vessel and the floor so that it just begins to slide on the surface is

(A) $\mu = \frac{a}{A}$

(B) $\mu = \frac{2a}{A}$

(C) $\mu = \sqrt{\frac{a}{A}}$

(D) $\mu = \sqrt{\frac{2a}{A}}$

Q.18

Which one of the following statements is correct? When a fluid passes through the narrow part of nonuniform pipe,

(A) its velocity and pressure both increase

(B) its velocity and pressure both decrease

(C) its velocity decreases but its pressure increases

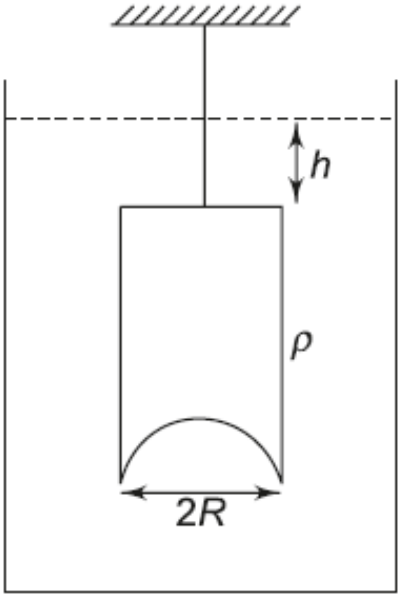
(D) its velocity increases but its pressure decreases

Q.19

A hemispherical portion of radius R is removed from the bottom of a cylinder of radius R. The volume of the remaining cylinder is V and its mass is M. It is suspended by a string in a liquid of density ρ where it stays vertical. The upper surface of the cylinder is at a depth h below the liquid surface. The force on the bottom of

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the cylinder by the liquid is (Fig.)



(A) Mg

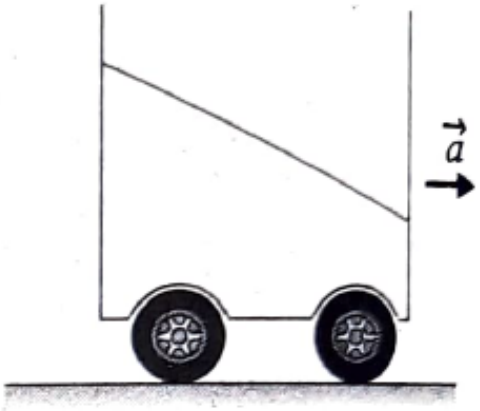
(B) $Mg - V\rho g$

(C) $Mg + \pi R^2 h\rho g$

(D) $\rho g\left(V + \pi R^2 h\right)$

Q.20

An open water tanker moving on a horizontal straight road has a cubical block of cork floating over its surface. If the tanker has an acceleration of a as shown, the acceleration of the cork w.r.t. container is



(A) Zero

(B) $\frac{a^2}{g}$

(C) $\frac{a}{y}\sqrt{g^2 - a^2}$

(D) a

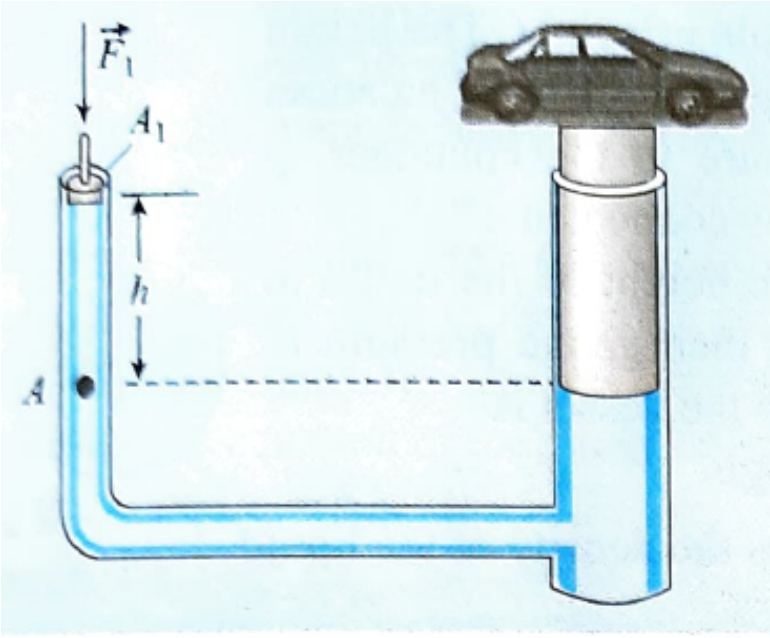
Q.21

Eight spherical rain drops of the same mass and radius are falling down with a terminal speed of 6 cms^{-1}. If they coalesce to form one big drop, what will be its terminal speed? Neglect the buoyancy due to air.

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Q.22

In the hydraulic car lift as shown in figure, the input piston on the left has a radius of $r_1 = 0.010\ \text{m}$ and a negligible weight. The output plunger on the right has a radius of $r_2 = 0.20\ \text{m}$. The combined weight of the car and the plunger is $20000\ \text{N}$. Suppose that the bottom surfaces of the piston and plunger are at the same level, so that $h = 0\ \text{m}$ in figure. What is the magnitude F_1 of the input force needed?



Q.23

A liquid flows steadily through two capillary tubes A and B connected in series. The lengths of A and B are L and $2L$ and their radii are r and $2r$. The ratio of pressure differences between the ends of tubes A and B is

Q.24

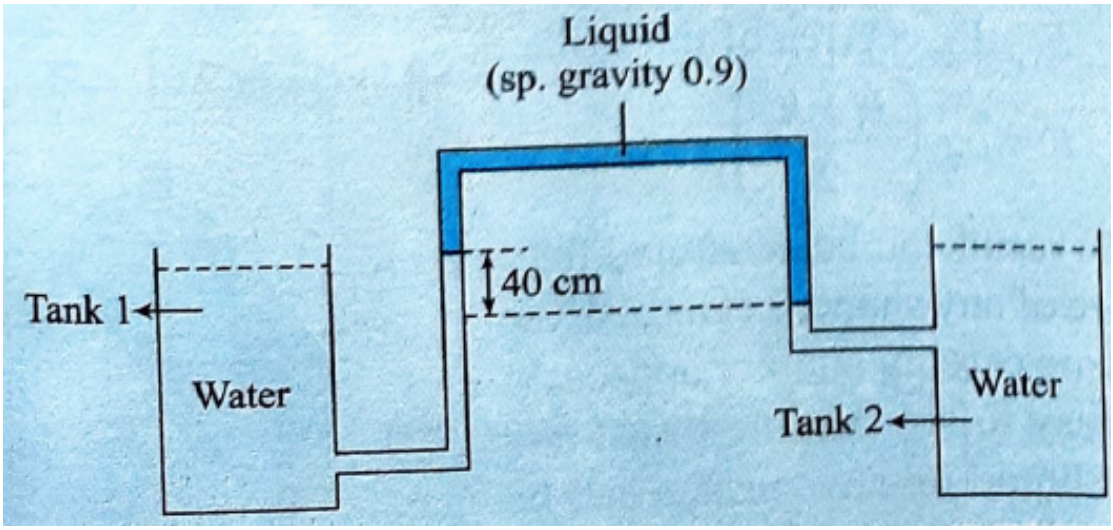
A drop of liquid of radius $R = 10^{-2}\ \text{m}$ having surface tension $S = \frac{0.1}{4\pi}\text{Nm}^{-1}$ divides itself into K identical drops. In this process the total change in the surface energy $\Delta U = 10^{-3}\ \text{J}$. If $K = 10^{\alpha}$ then the value of α is

Q.25

An inverted U-tube manometer shown in figure is used to measure the difference in water level between the

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two tanks. Calculate this difference for the conditions indicated.



Q.26

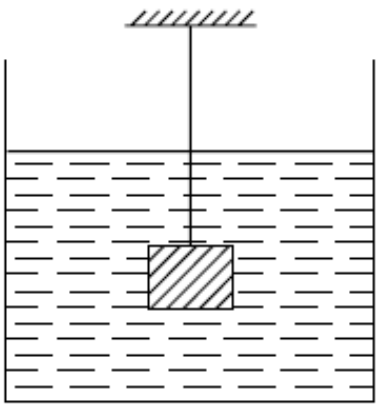
A cylindrical vessel of a large crosssectional area has two tiny holes A and B and lies on a horizontal floor. Hole A is at a depht h_1 below the free surface of liquid in the vessel and hole B is at a height h_2 above the bottom of the vessel. When the holes are unplugged, the streams emerging from A and B strike the floor at the same point. The ratio h_1/h_2 is

Q.27

A cylindrical tank having cross-sectional area A is filled with water to a height of $2.0\ \mathrm{m}$. A circular hole of cross-sectional area a is opened at a height of $75\ \mathrm{cm}$ from the bottom. If $\frac{a}{A} = \sqrt{0.2}$, , the velocity with which water emerges from the hole is $\left(g = 9.8\ \mathrm{ms}^{-2}\right)$

Q.28

A cubical metal block of edge $5\ \mathrm{cm}$ is suspended from a support by a massless string and immersed in water in a beaker as shown in Fig. The relative density of metal is 7. Find the tension in the string when the whole system moves up with an acceleration of $2\ \mathrm{ms}^{-2}$



Q.29

What will be the length of mercury column in a barometer tube, when the atmospheric pressure is $75\ \mathrm{cm}$ of mercury and the tube is inclined at an angle of 60° to the vertical?

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Q.30

What is the absolute pressure on a swimmer $10\ \mathrm{m}$ below the surface of a lake? Take atmospheric pressure $1 \times 10^5\ \mathrm{N/m^2}$

Answers & Solutions

Q.1 Answer:

$\frac{1}{3}A\rho gh$

Solution:

The Area of the lamina is A and the height is h which is immersed in a liquid of the density ρ in a vertical plane with its base on the surface of the liquid.

The thrust on the lamina = pressure at centroid $\times$ Area

$= \frac{\mathrm{hPg}}{3} \times \mathrm{A} = \frac{1}{3}\ \mathrm{APgh}$

Q.2 Answer:

$\sqrt{\frac{gh}{2}}$

Solution:

When 75% of the liquid has drained out, the height of the liquid in the container will be $h' = h/4$.

For this height,

Velocity of liquid flowing out $(v') = \sqrt{2gh'} = \sqrt{2g \times \frac{h}{4}} = \sqrt{\frac{gh}{2}}$, which is choice (a).

Q.3 Answer:

1000

Solution:

Let the volume of the block be $V\ \mathrm{m}^3$.

Volume of block under liquid $= \frac{4V}{5}\ \mathrm{m}^3$

$\therefore$ **Volume of liquid displaced $= \frac{4V}{5}\ \mathrm{m}^3$**

Now let the density of the liquid be $\rho \mathrm{kgm}^{-3}$.

Mass of liquid displaced

= (volume of liquid displaced) $\times$ (density of liquid)

$= \frac{4V}{5}\rho \mathrm{kg}$

Weight of liquid displaced $= \frac{4V}{5} \times \rho \times g$ newton

Relative density of wood $= 0.8$

$$\therefore \quad \text{Density of wood } = 0.8 \times 1000$$
$$= 800\ \mathrm{kg\ m^{-3}}$$

$\therefore$ **Mass of the block** $= 800 \times V\ \mathrm{kg}$

From the law of flotation,

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Weight of block = weight of liquid displaced

or $800 \times V \times g = \frac{4V}{5} \times \rho \times g$

$$\rho = 800 \times \frac{5}{4}$$

$$\text{Or } = 1000 \text{ kg m}^{-3}$$

Hence the correct choice is (b).

Q.4 Answer:

$\frac{V}{n}\left(\frac{R}{r}\right)^2$

Solution:

Cross-sectional area of tube $(A) = \pi R^2$**. Crosssectional area of each hole** $= \pi r^2$**. Therefore, crosssectional area of** n **holes** $(a) = \pi n r^2$**. If** v **is the speed of ejection of the liquid through the holes, then from the continuity of flow, we have**

$$av = AV$$

$$v = \frac{AV}{a} = \frac{\pi R^2 V}{\pi n r^2} = \frac{V}{n}\left(\frac{R}{r}\right)^2$$

Hence the correct choice is (d).

Q.5 Answer:

$\left|2P_0Rh + R\rho gh^2 - 2RT\right|$

Solution:

Let us find the force exerted on water in the right side of section $ABCD$**. This force** $F =$ **force** f_1**, due to pressure on the left side of section** $ABCD+$ **force** f_2 **due to surface tension. Now**

$$f_1 = \text{ pressure at the center } \times \text{ area of } ABCD$$

$$= \left(P_0 + \tfrac{h}{2}\rho g\right) \times 2Rh \text{ directed towards right}$$

$$f_2 = \text{ surface tension } \times \text{ length } AB$$

$$= T \times 2R = 2RT \text{ directed toward left}$$

$$\therefore \quad F = \left(P_0 + \frac{h}{2}\rho g\right) \times 2Rh - 2RT$$

$$= \left[2P_0Rh + R\rho gh^2 - 2RT\right]$$

Q.6 Answer:

8 ms^{-1}

Solution:

From the equation of continuity of flow, the velocities v_1 **and** v_2 **of the liquid at points where the areas of cross-section of the pipe are** a_1 **and** a_2 **respectively are related as**

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$a_1 v_1 = a_2 v_2$

$\therefore \quad v_2 = \frac{a_1 v_1}{a_2} = \frac{\pi r_1^2 v_1}{\pi r_2^2}$

$= v_1 \left(\frac{r_1}{r_2}\right)^2$

$= 2 \times \left(\frac{6}{3}\right)^2 = 8 \text{ ms}^{-1}$

Hence the correct choice is (c).

Q.7 Answer:

lower in the front side

Solution:

Due to frictional force (which acts in a direction opposite to the direction of acceleration) on the rear face, the pressure in the rear side will be increased. Hence the pressure in the front side will be lowered. Hence the correct choice is (b).

Q.8 Answer:

$P_0 + \frac{mg}{a}$

Solution:

The forces acting on the piston are

(i) force $P_0 a$ due to atmospheric pressure acting downwards

(ii) weight mg of the piston acting downwards

(iii) force Pa due to liquid below the piston.

Since the piston is in equilibrium,

$Pa = P_0 a + mg$

$\Rightarrow \quad P = P_0 + \frac{mg}{a}$

Q.9 Answer:

$\frac{R^2}{4h}$

Solution:

Velocity of efflux $(v) = \sqrt{2gd}$. Time taken

$t = \sqrt{\frac{2h}{g}}$

$\therefore$ Range $R =$ velocity $\times$ time $= vt = \sqrt{2gd} \times \sqrt{\frac{2h}{g}}$

$= \sqrt{4dh}$

or $R^2 = 4dh$ or $d = \frac{R^2}{4h}$, which is choice (c).

Q.10 Answer:

$\frac{\sigma - \rho_2}{\rho_1 - \sigma}$

Solution:

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Mass of the body is $m = \sigma AL$, where A is the cross-sectional area of the cylinder. Weight of cylinder is $W = mg = \sigma ALg$.

Weight of volume of liquid of density ρ_1 displaced by the length L_1 is

$$W_1 = \rho_1 AL_1 g$$

Weight of volume of liquid of density ρ_2 displaced by length L_2 is

$$W_2 = \rho_2 AL_2 g$$

According to the law of floatation, the weight of the cylinder must be equal to the total weight of the liquids displaced, i.e.

$$W = W_1 + W_2$$

$$\sigma ALg = \rho_1 AL_1 g + \rho_2 AL_2 g$$

$$\Rightarrow \quad \sigma L = \rho_1 L_1 + \rho_2 L_2 \quad \text{(i)}$$

$$\text{Now} \qquad L_1 + L_2 = L \quad \text{(ii)}$$

From (ii), $L_2 = L - L_1$. Using this in Eq. (i),

$$\sigma L = \rho_1 L_1 + \rho_2 (L - L_1)$$

$$\Rightarrow \quad L_1 = \left(\frac{\sigma - \rho_2}{\rho_1 - \rho_2}\right) L$$

Similarly, using $L_1 = L - L_2$ in Eq. (i), we get

$$L_2 = \left(\frac{\rho_1 - \sigma}{\rho_1 - \rho_2}\right) L$$

$$\therefore \frac{L_1}{L_2} = \frac{\sigma - \rho_2}{\rho_1 - \sigma}$$

Q.11 Answer:

$\frac{1}{4}(3n+1)$

Solution:

Total upward buoyant force on the block is

$$U = \text{buoyant force due to oil} + \text{buoyant force due to water}$$

$$= \rho_0 (l - x) l^2 g + \rho_w x l^2 g$$

For floating U = weight of the block $= \rho_b l^3 g$

$$\therefore \quad \rho_b l^3 g = \rho_0 (l - x) l^2 g + \rho_{\text{w}} x l^2 g$$

$$\Rightarrow \quad \frac{\rho_b}{\rho_w} = \frac{1}{l}\left[\frac{\rho_0}{\rho_w}(l - x) + x\right]$$

$$= \frac{1}{l}[n(l - x) + x]$$

Putting $x = \frac{l}{4}$**, we get**

$$\frac{\rho_b}{\rho_w} = \frac{1}{4}(3n+1)$$

So the correct choice is (c).

Q.12 Answer:

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$\frac{2\sigma}{r}$

Solution:

For an air bubble is formed inside a soap solution, there is only one surface in contact with soap solution. Hence the correct choice is (b).

Q.13 Answer:

$4\pi\sigma R^2$

Solution:

Volume of big drop $= \frac{4}{3}\pi R^3$. **If** r **is the radius of each tiny drop, the total volume of eight drops =** $8 \times \frac{4\pi}{3} r^3 = \frac{32}{3}\pi r^3$. **Since the total volume remains unchanged, we have**

$\frac{32}{3}\pi r^3 = \frac{4}{3}\pi R^3$

which gives $r = \frac{R}{2}$. **Now, surface area of big drop =** $4\pi R^2$.

Total surface area of eight drops $= 32\pi r^2 = 8\pi R^2$

$(\because r = R/2)$

$\therefore$ **Increase in surface area** $= 8\pi R^2 - 4\pi R^2 = 4\pi R^2$

$\therefore$ **Work done = surface tension** $\times$ **increase in surface area** $= 4\pi\sigma R^2$

Hence the correct choice is (a).

Q.14 Answer:

r^5

Solution:

The terminal velocity is

$v_t = \frac{2}{9}\frac{(\rho-\sigma)r^2 g}{\eta}$

The rate of production of heat (or power dissipated) is given by

$$P = f v_t, \text{ where } f = 6\pi\eta r v_t$$

$$\therefore \quad P = 6\pi\eta r v_t \times v_t$$

$$= 6\pi\eta r \left[\frac{2}{9}\frac{(\rho-\sigma)r^2 g}{\eta}\right]^2$$

$$= \left[\frac{8\pi g^2}{27\eta}(\rho-\sigma)^2\right] r^5$$

i.e. $P \propto r^5$

Q.15 Answer:

$\frac{F}{4}$

Solution:

$$F = 6\pi\eta r v$$

$$F' = 6\pi\eta\left(\frac{r}{2}\right) \times \left(\frac{v}{2}\right) = \frac{1}{4} \times 6\pi\eta r v = \frac{F}{4}$$

So the correct choice is (c).

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Q.16 Answer:

10 g

Solution:

Mass of water in first tube is

$m = \pi r^2 h \rho$

Now, surface tension $\sigma = \frac{h\rho g r}{2} = \frac{h'\rho g r'}{2}$

where h' **is the height to which water rises in the second tube and** r' **its radius. Since** $r' = 2r, h' = h/2$ **Therefore, the mass of water in the second capillary tube is**

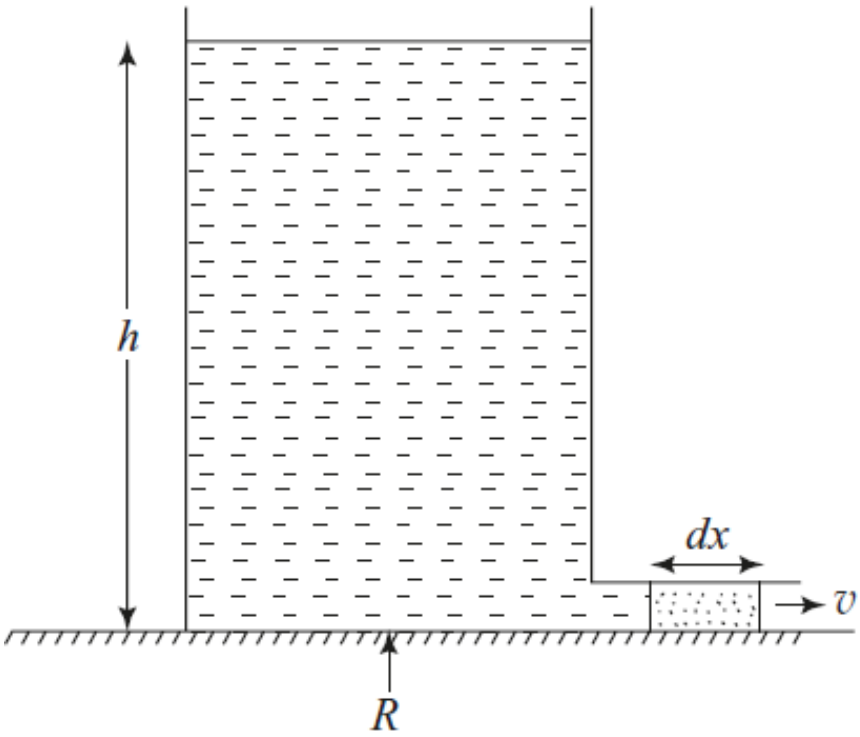
$$m' = \pi r^2 h' \rho = \pi (2r)^2 \frac{h}{2} \rho$$
$$= 2\pi r^2 h \rho = 2m = 2 \times 5 = 10 \text{ g}$$

Hence the correct choice is (c).

Q.17 Answer:

$\mu = \frac{2a}{A}$

Solution:



velocity of efflux is $v = \sqrt{2gh}$

From Newton's second law, the backward force exerted on the vessel by the emerging stream of liquid is

$F = \frac{dp}{dt} = \frac{d}{dt}(mv) = v\frac{dm}{dt} \quad (\because v = \text{ constant })$

Let dx **be the distance moved by the liquid in time** dt**. Volume of liquid flowing out in time** $dt = adx$**. Mass of liquid flowing out in time** $dt = \rho$ **a** dx**. Therefore, rate of flow of mass is**

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$$\frac{dm}{dt} = \rho a \frac{dx}{dt} = \rho a v$$

$$\therefore F = v \times \rho a v = \rho a v^2 = \rho a \times 2gh = 2\rho g h a$$

Frictional force $f = \mu R = \mu M g$

$(M \quad = \text{ mass of liquid in vessel })$

$= \mu \times \rho A \quad h \times g$

The vessel will just begin to slide if $F = f$**, i.e.**

$$2\rho g h a = \mu \rho A h g$$

$$\Rightarrow \mu = \frac{2a}{A}$$

Q.18 Answer:

its velocity increases but its pressure decreases

Solution:

Consider two points A **and** B **in the pipe at the same horizontal level. At point** A**, let** a_1 **be the area of cross-section of the pipe,** v_1 **the velocity of fluid flow and** p_1 **the fluid pressure. Let** a_2**,** v_2 **and** p_2 **be the corresponding quantities at point** B**. Then from Bernoulli's theorem, we have**

$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$

or $\quad (p_1 - p_2) = \frac{1}{2}\rho\left(v_2^2 - v_1^2\right)$ **(i)**

If $a_2 < a_1$**, it follows from continuity equation** $a_1 v_1 = a_2 v_2$**, that** $v_1 > v_2$**. From Eq. (i), it follows that** $p_2 < p_1$**. Hence, at the narrow part of the pipe, the fluid velocity increases but its pressure decreases, which is choice (d).**

Q.19 Answer:

$\rho g\left(V + \pi R^2 h\right)$

Solution:

From Archimedes' principle, $F_b - F_t =$ **upthrust = weight of volume** V **of the liquid. Here** F_b **and** F_t **denote the force exerted by the liquid on the bottom and the top of the cylinder respectively.**

Upthrust = $\rho \mathrm{Vg}$ **and** $F_t = \left(\pi R^2 h\right)\rho g$**. Hence**

$F_b = \left(\pi R^2 h\right)\rho g + \rho V g = \rho g\left(\pi R^2 h + V\right)$

Hence the correct choice is (d).

Q.20 Answer:

Zero

Solution:

After breaking the force into components

And balancing them we get

$\mathrm{mat}_1 = \mathrm{mg\,Sin}\,\theta - \mathrm{maCos}\,\theta$

we know that for water surface $\tan\theta = \frac{\mathrm{a}}{\mathrm{g}} \rightarrow \quad \mathrm{a_{n1}} = 0$

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Q.21 Answer:

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Solution:

Let r be the radius of each small drop. The volume of each drop $= \frac{4}{3}\pi r^3$. Let R be the radius of the big drop formed when 8 small drops coalesce. The volume of the big drop will be 8 times that of the small drop. Hence

$$\frac{4}{3}\pi R^3 = 8 \times 4\pi \frac{r^3}{3}$$
$$R^3 = 8r^3$$
$$R = 2r$$

i.e. the radius of the big drop is twice the radius of each small drop. Now, if the buoyancy of the drop due to air is neglected, the terminal speed of the small drop is given by

$v_t = \frac{2}{9}\frac{\rho r^2 g}{\eta}$ (i)

The terminal speed of the big drop will be

$V_t = \frac{2}{9}\frac{\rho R^2 g}{\eta}$ (ii)

Dividing (i) and (ii), we have

$$\frac{V_t}{v_t} = \frac{R^2}{r^2} = \frac{(2r)^2}{r^2} = 4$$
$$V_t = 4v_t = 4 \times 6 = 24 \text{ cm s}^{-1}$$

Q.22 Answer:

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Solution:

The output force should have a magnitude of $F_2 = 20000$ N to supports the car. Initially the piston and the plunger are at the same height, so the fluid pressure at the piston is equal to the fluid pressure at the plunger $\frac{F_1}{A_1} = \frac{F_2}{A_2}$ where A_1 is the area of the piston and A_2 is the area of the plunger.

Therefore, the magnitude of the force $F_1 = F_2\left(\frac{A_1}{A_2}\right)$

Using $A = \pi r^2$ for the circular areas of the piston and plunger.

We find that, $F_1 = F_2\left(\frac{A_1}{A_2}\right) = F_2\left(\frac{\pi r_1^2}{\pi r_2^2}\right) = 20000\left(\frac{0.01}{0.2}\right)^2 = 50$ N

Q.23 Answer:

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Solution:

The volume of a liquid flowing per second is through a capillary tube is given by

$Q = \frac{\pi p r^4}{8\eta L}$

where η is the coefficient of viscosity of the liquid and p is the pressure difference between the ends of the tube. Since the liquid does not accumulate anywhere in the tube, $Q_1 = Q_2$ or

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$$\frac{\pi p_1 r_1^4}{8\eta L_1} = \frac{\pi p_2 r_2^4}{8\eta L_2}$$

$$\Rightarrow \quad \frac{p_1}{p_2} = \left(\frac{L_1}{L_2}\right) \times \left(\frac{r_2}{r_1}\right)^4$$

$$= \left(\frac{L}{2L}\right) \times \left(\frac{2r}{r}\right)^4$$

$$= \frac{1}{2} \times 2^4 = 2^3 = 8$$

Q.24 Answer:

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Solution:

$\frac{4}{3}\pi R^3 = K\frac{4}{3}\pi r^3, R = 10^{-2}$

$r = R \times k^{\frac{-1}{3}}$

$\mathrm{d}A = \left(k \times 4\pi r^2\right) - 4\pi R^2$

$dA = 4\pi R^2 \left[\mathrm{k}^{\frac{1}{3}} - 1\right]$

$\Delta \mathrm{U} = \mathrm{T} \times \mathrm{dA}$

$10^{-3} = \frac{0.1}{4\pi} \times 4\pi \mathrm{R}^2 \left[\mathrm{k}^{\frac{1}{3}} - 1\right]$

$10^2 = \left[\mathrm{k}^{\frac{1}{3}} - 1\right]$

$\mathrm{k} \approx 10^6$

$\alpha = 6$

Q.25 Answer:

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Solution:

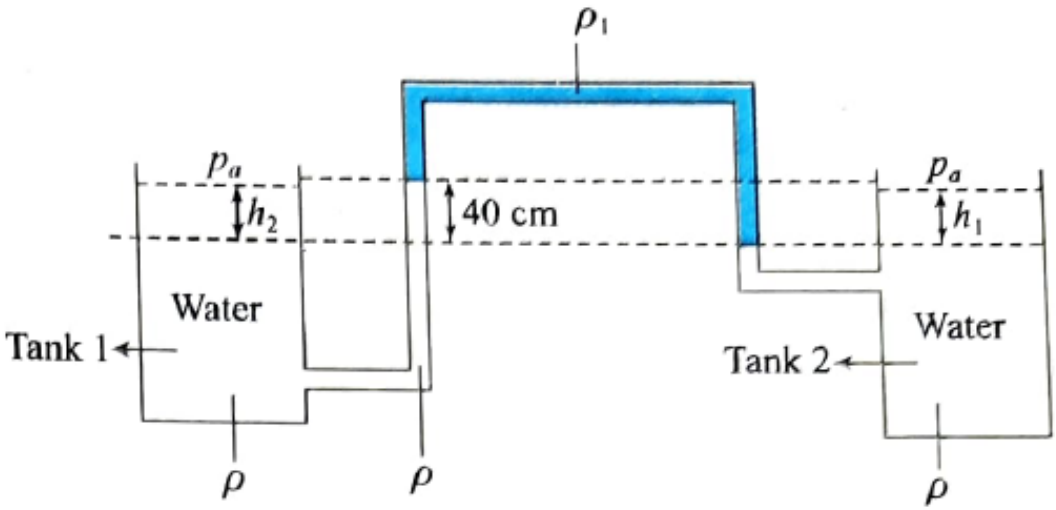
The inverted U-tube manometer is used for finding the difference of water level in two tanks. Draw a horizontal

straight line passing through the lower level and upper level liquid in the manometer as shown in figure. Starting from top of tank 2 and moving through the tube and reaching the top of tank 1, the manometric equation can be written as

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$p_a + h_1\rho g - 40\rho_1 g + 40\rho g = p_a + h_2\rho g$

$h_2\rho g - h_1\rho g = 40\rho g - 40\rho_1 g$



But we are given, $\rho_1 = 0.9\rho$

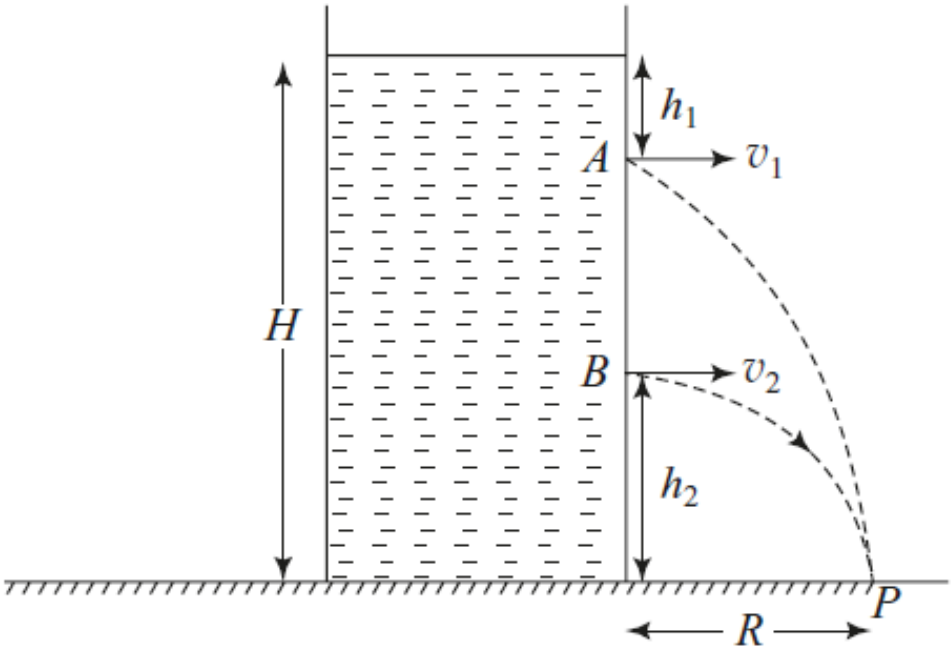
$\therefore \qquad (h_2 - h_1)\,\rho g = 40\rho g - 36\rho g$

or $\quad h_2 - h_1 = 4 \text{ cm}$

Q.26 Answer:

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Solution:



Velocity of efflux is

$v_1 = \sqrt{2gh_1}$

$v_2 = \sqrt{2g\,(H - h_2)}$

The times taken by the liquid emerging from A **and** B **to hit the floor are**

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$$t_1 = \sqrt{\frac{2\left(H - h_1\right)}{g}}$$

$$t_2 = \sqrt{\frac{2h_2}{g}}$$

Since the two streams strike the floor at the same point P, $R_1 = R_2$ or $v_1 t_1 = v_2 t_2$ or

$$\sqrt{2gh_1} \times \sqrt{\frac{2\left(H - h_1\right)}{g}} = \sqrt{2g\left(H - h_2\right)} \times \sqrt{\frac{2h_2}{g}}$$

$$\Rightarrow h_1\left(H - h_1\right) = \left(H - h_2\right) h_2$$

$$\Rightarrow \left[H - \left(h_1 + h_2\right)\right]\left[h_1 - h_2\right] = 0$$

which gives $H = h_1 + h_2$ (which is not possible) and $h_1 = h_2$ So the correct choice is (d).

Q.27 Answer:

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Solution:

Let v be the velocity of efflux and V be the velocity with which the water level in the tank falls. From the equation of continuity of flow, we have $av = AV$ or

$V = \frac{av}{A}$

From Bernoulli's theorem, we have

$P_0 + \frac{1}{2}\rho V^2 + \rho g H = P_0 + \frac{1}{2}\rho v^2 + \rho g h$

which gives $v^2 = V^2 + 2g(H - h)$ (2)

Using (1) in (2), we get

$V^2 = \frac{2g(H-h)}{1-\left(\frac{a}{A}\right)^2} = \frac{2\times 9.8\times(2.0-0.75)}{1-0.2} = 25$

which gives $v = 5.0\ \mathrm{ms}^{-1}$

Q.28 Answer:

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Solution:

In this case $g_{\text{eff}} = g + a$

$$\therefore \quad T' = W' - U' = \sigma V g_{\text{eff}} - \rho V g_{\text{eff}}$$
$$= (\sigma - \rho) V (g + a)$$
$$= (7000 - 100) \times \left(1.25 \times 10^{-4}\right) \times (10 + 2)$$
$$= 9.0\ \mathrm{N}$$

Q.29 Answer:

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Solution:

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Given:

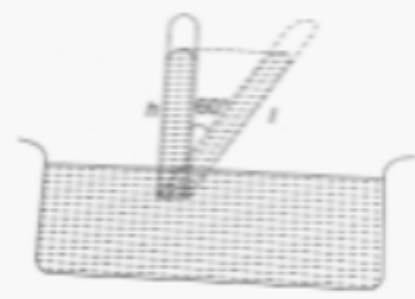
$\mathrm{P} = 7\mathrm{scm}$ of Hg

Pressure is develop due to vertical hight in the barometer bence atmospheric pressure, $P =$

$\mathrm{lcos}\,\theta = \mathrm{lcos}\,60^\circ = 1/2$

$1 = 2\mathrm{P} = 2 \times 75$ cm

$1 = 150$ cm



Q.30 Answer:

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Solution:

The formula used to calculate the absolute pressure, the pressure that varies with the depth of the fluid is given as follows:

$P = P_0 + \rho g h$

We need to substitute the given parameters in the above equation as,

$P_S = P_A + \rho g h$

In the question we are provided that the pressure exerted by the air is equal to 1.01×10^5 Pa**, the depth at which the swimmer is swimming is** 10 m **and the density of the vater is** $1000\ \mathrm{kg/m^3}$**.**

Upon substituting the above values in the above equation, we get,

$P_S = 1.01 \times 10^5 + 1000 \times 10 \times 10$

On calculating these values we get,

$P_S = 2 \times 10^5$ Pa

Now we need to perform the unit conversion, that is, the value of the pressure we have obtained is in terms of the Pascal, but in the options, the unit used is atm, so we will convert the unit to the same.

The conversion factor from Pascal to atmosphere is, 1 Pa $= 10^{-5}$ atm

Using this relation, we will convert the unit of the pressure value obtained.

So, we get,

$P_S = \left(2 \times 10^5\right) \times 10^{-5}$

So on doing the calculation we have,

$P_S = 2$ atm

Hence the pressure on the swimmer is 2 atm**.**

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